

$\therefore \frac{(x+y)^2}{4} \leq xy \Rightarrow (x+y)^2 \leq 4xy \Rightarrow x^2 + 2xy + y^2 \leq 4xy \Rightarrow x^2 - 2xy + y^2 \leq 0$
 $\Rightarrow (x-y)^2 \leq 0$. $(x-y)$ is a real no. The square of any real no. must be non-negative. $\therefore (x-y) = 0 \Rightarrow \boxed{x=y}$. This is a contradiction to the fact that x and y are two positive distinct reals, i.e. $\boxed{x \neq y}$.

$$\therefore \frac{(x+y)}{2} > \sqrt{xy} \quad \square$$

Example 15: Suppose that two people play a game taking turns removing one, two, or three stones at a time from a pile that begins with 15 stones. The person who removes the last stone wins the game. Show that the first player can win the game no matter what the second player does.

Solution: We have to show that the first player can always remove the 15th stone no matter what strategy the second player follows.
 Suppose, the first player removes the 11th stone. Now, if the second player takes two stones, the first player can remove the remaining two stones and win. If the second player takes 3 stones, the first player takes the 15th stone and win. If the second player takes 1 stone, the first player takes the remaining 3 stones and win. $\therefore$ Our strategy must be designed in such a way that the first player always removes the 11th stone.

The first player will remove 3 stones in the begin. There are 8 more stones left to remove in order for the first player to remove the 11 stone no.
 $\therefore$ We must remove the 8 more stones in an even no. of moves.
 If 2nd player removes 3 stones, first player removes 1 stone.
 If 2nd player removes 2 stones, first player removes 2 stones.
 If 2nd player removes 1 stone, 1st player removes 3 stones.